

# Adaptive Trial Design for the RAS-Inhibitor Era: Combinations, Resistance, and Biomarker-Guided Therapy

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# Today's argument

**New RAS therapies in PDAC need new kinds of trials.**

The statistical methods exist to **optimize combination doses**.

The statistical methods exist to **adapt trials to emerging resistance**.

**The challenge is making these designs practical enough to launch.**

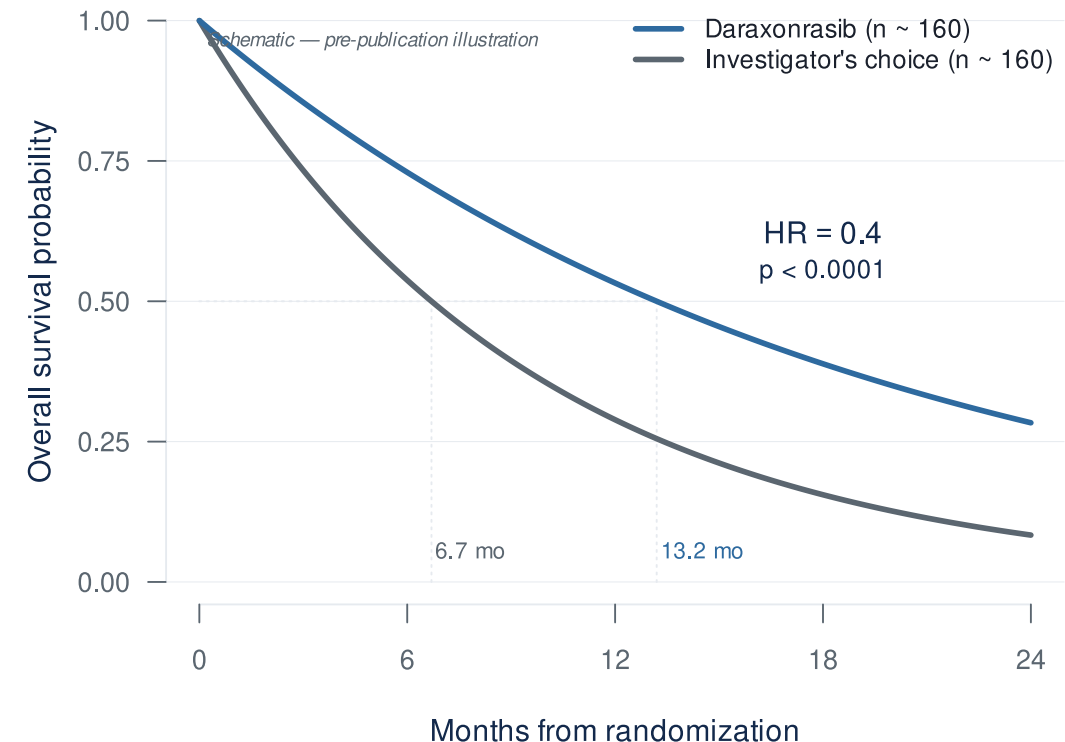
# The clinical case

*Why combinations, and why now?*

# The RAS revolution has arrived for PDAC

A major survival signal from direct RAS inhibition in PDAC.

- **RASolute 302 (2L mPDAC, *NEJM* 2026)<sup>1</sup>:**
  - **Median OS: 13.2 vs 6.7 months**
  - **HR 0.40, 60% reduction in risk of death ( $p < 0.0001$ )**
- Daraxonrasib (RMC-6236), oral, multi-selective RAS(ON) inhibitor
- ASCO 2026 Plenary, May 31 (Wolpin)



# ...but monotherapy has a narrow tolerability margin

*RMC-6236-001 (PDAC cohort, NEJM 2026)<sup>2</sup>*



**48% of monotherapy patients require dose modification at 300 mg, a strong signal of limited tolerability margin.** Combinations can layer **overlapping, non-additive** toxicity that single-agent data don't reliably predict, which is why the combination needs its own dose-finding.

# Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK pathway activation** on PDAC cells, through cell-surface upregulation<sup>3</sup> and recurrent genomic alterations<sup>4</sup>, the mechanistic case for biomarker-driven, resistance-preemptive combinations
- **1L combination cohort** (daraxonrasib 200 mg + GnP, n=40)<sup>5</sup>:
  - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



RTK pathway activation is simultaneously a resistance mechanism and a rational combination target

*The same adaptation that creates resistance may create a therapeutic vulnerability.*

# The dose-finding problem

*Why MTD only may not be best for RAS combinations, and what replaces it.*

# Why combinations break the rules you grew up on

- **Rivière's review of 162 combination Phase I trials: 88% used 3+3** even when both drugs were escalated<sup>6,7</sup>.
- **Why 3+3 doesn't fit combinations.** Built for single-agent cytotoxics<sup>8</sup>, it assumes monotonic dose-toxicity, Cycle-1-observable DLTs, and a single dose dimension, combinations strain all three (e.g., **immune toxicities often emerge weeks later, outside the DLT window**).
- **The two-drug dose map is never built.** In practice: pick A's dose, fix it, add B. Two monotherapy escalations run in series; the joint efficacy/toxicity behavior across (A, B) pairs (the "joint surface") is never characterized.

*Two-drug dose map: how efficacy and toxicity change across every (A, B) dose pair, not just along one drug's axis.*



# The Amgen case: sotorasib + pembrolizumab

*A real-world example of why combination toxicity can surprise us.*

Same sponsor, two different designs:

- **Monotherapy (CodeBreakK 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied, joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window<sup>9</sup>

What MTD-only dose-finding was never built to handle, all at once: joint surface unsearched, non-additive toxicity, late-onset toxicity.

# Why MTD-only dose-finding doesn't fit targeted agents and immunotherapies



- **Pembrolizumab**, efficacy plateaus at the lowest dose tested; MTD never reached<sup>10</sup>
- **CAR-T**, efficacy *rises then falls* (umbrella); high doses drive severe CRS/ICANS, not benefit
- **A RAS combination spans both shapes.** Neither rewards titrating to toxicity.

# The cost of MTD-only dose-finding

- Sotorasib (KRAS G12C, first-in-class, approved 2021)<sup>11</sup>:
  - 960 mg selected as the **highest dose tested**, no DLTs, a more-is-better default, not an MTD
- **MTD  $\neq$  optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions<sup>12</sup>

**240 mg  $\approx$  960 mg on PFS**

*FDA-required post-marketing comparison: similar PFS at **one-quarter** the exposure<sup>13</sup>.*

*Same drug class as daraxonrasib.*

# Project Optimus: the regulatory ground has shifted

- FDA final guidance, 2024<sup>14</sup>
- Sponsors must **study more than one dose** and justify on efficacy and safety
- **Maximum Tolerated Dose → Optimal Biological Dose (OBD)**, a dose optimized on benefit-risk

**OBD** = the dose that **optimizes the efficacy–toxicity trade-off**, chosen by weighing the value of a response against the cost of toxicity, not by escalating until DLTs cap the dose<sup>15,16</sup>.

**Thall et al. *Clin Trials* 2024:** conventional Phase I designs cannot reliably identify safe and effective doses<sup>17</sup>.

# How OBD selection actually works

Instead of escalating until toxicity, choose the dose with the best benefit-risk profile.

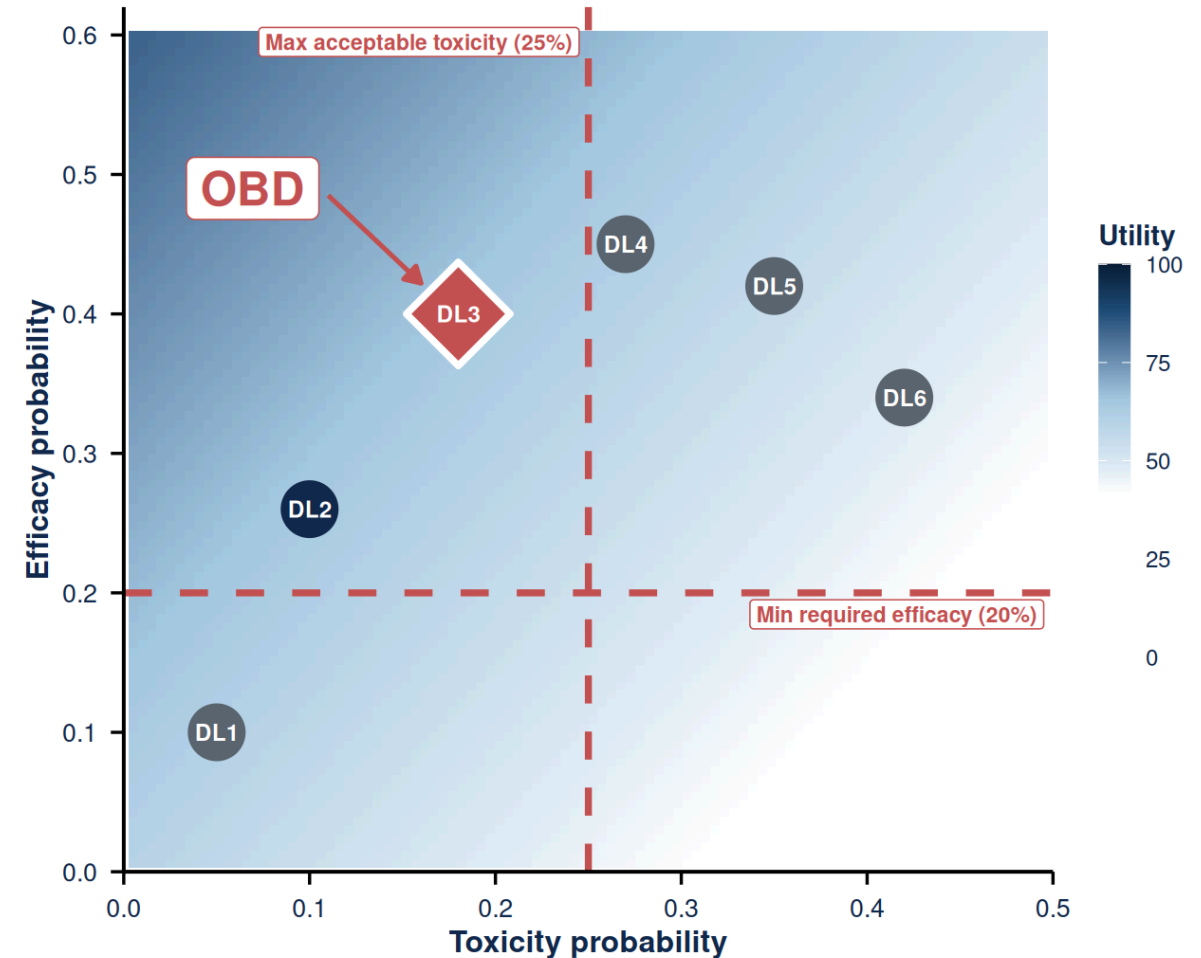
*“How much efficacy would you trade for additional toxicity?”*

The team writes that utility table down up front. The design then searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Elicited utility table (Zhou 2019 framework)<sup>15</sup>:

|                 | TOX       | NO TOX     |
|-----------------|-----------|------------|
| <b>Response</b> | <b>40</b> | <b>100</b> |
| No response     | <b>0</b>  | 60         |

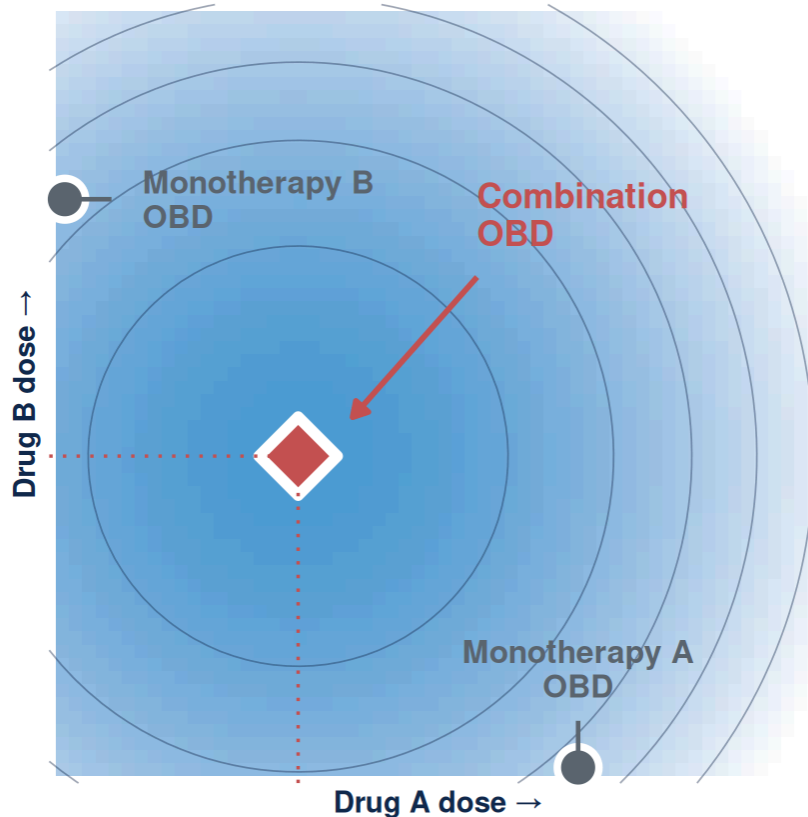
*A tolerated non-response (60) is preferred to a toxic response (40).*



*OBD = admissible dose (inside red lines) with the highest utility.*

# Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



- The combination OBD can sit **below** both **single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **tolerability margin is already limited** at the monotherapy dose

**Fix what you know. Search what you don't.**

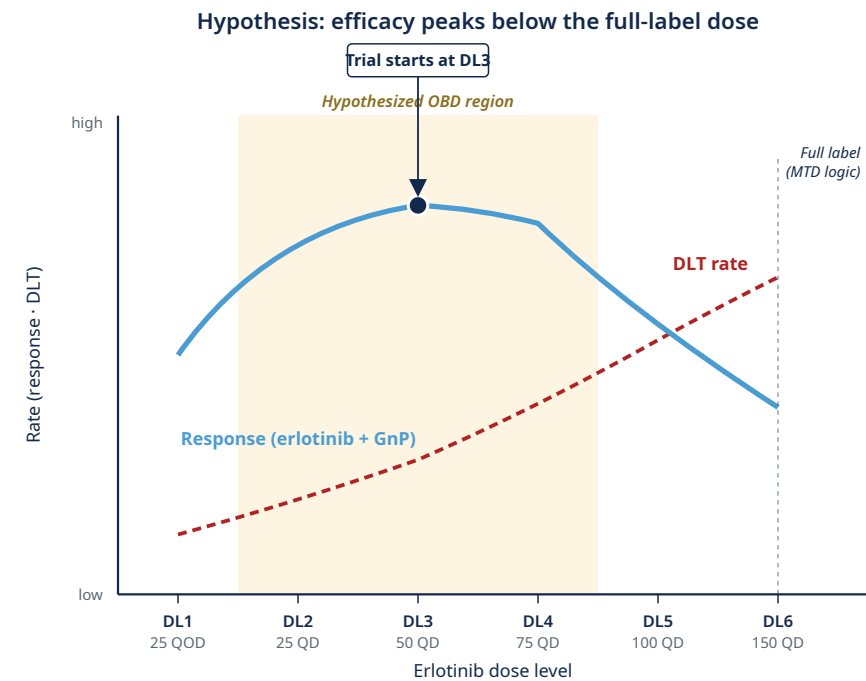
*RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface<sup>5</sup>.*

*Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.*

# Finding the right erlotinib dose in basal-like PDAC

- **Clinical question.** Could *lower*-dose erlotinib work *better* than full-dose erlotinib in basal-like PDAC, when combined with GnP?
- **Design.** First find the safe dose range, then choose the dose with the best response/toxicity trade-off<sup>15,18</sup>.
- **Why it matters.** This is an OBD-below-MTD question; 3+3 cannot answer it.

**Trial:** LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by PurlIST<sup>19</sup> · combination erlotinib + biweekly GnP · OBD-below-MTD hypothesized from prior data<sup>20</sup> · **up to 52 patients; ~28 expected at the OBD.**

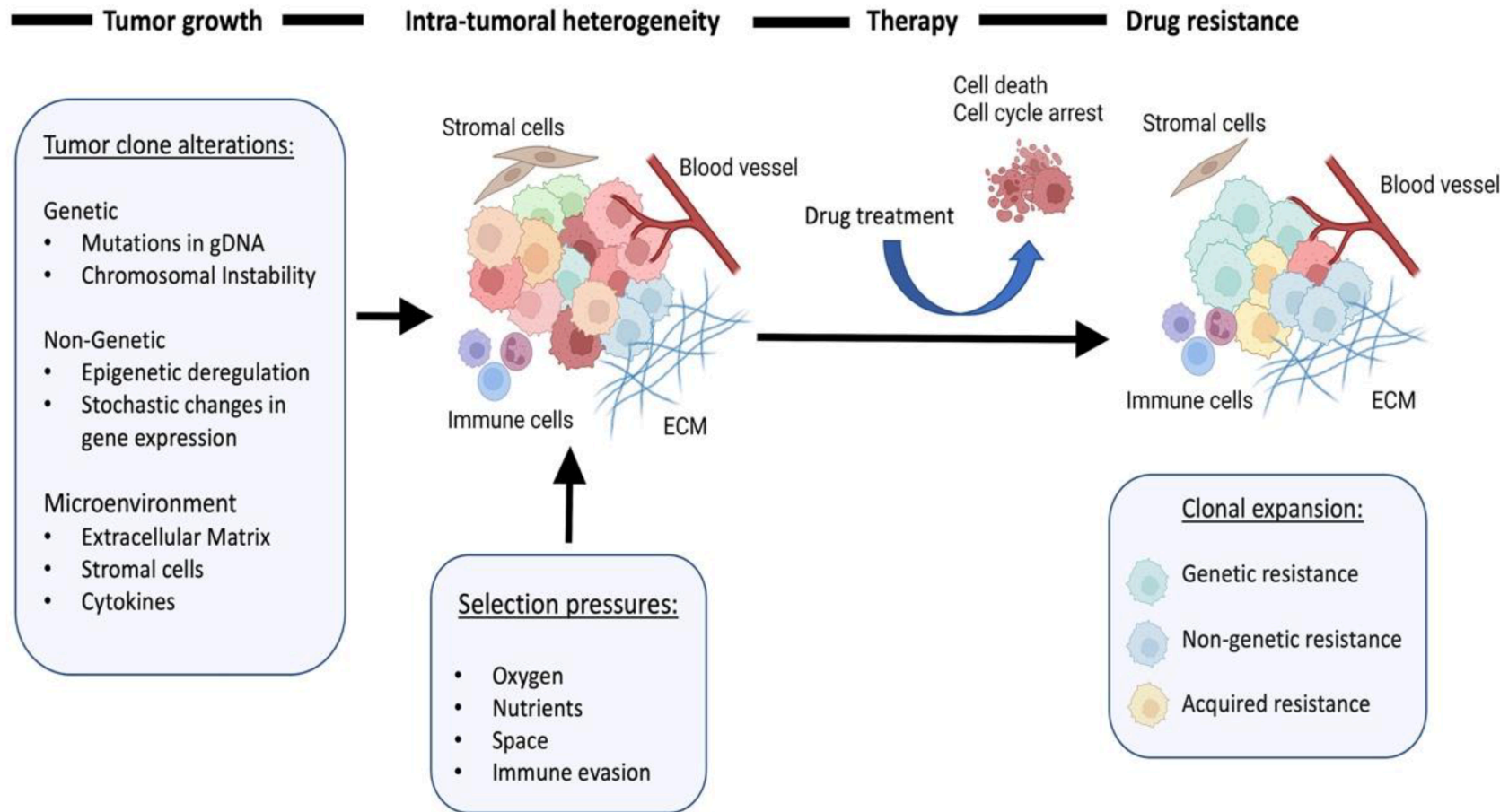


# Resistance is the next problem

*Phase 2 onward: each patient is a moving target*



# Resistance is a dynamic process



Resistance mechanisms emerge under RAS-inhibitor pressure

**Daraxonrasib monotherapy in mPDAC (n=44, paired ctDNA): 59% show RAS-pathway resistance at progression, KRAS amplification in 36%**<sup>22</sup>

Adapted from Karagiannis & Rampias, Cancers 2022<sup>21</sup>.

# Why classical trial designs miss resistance

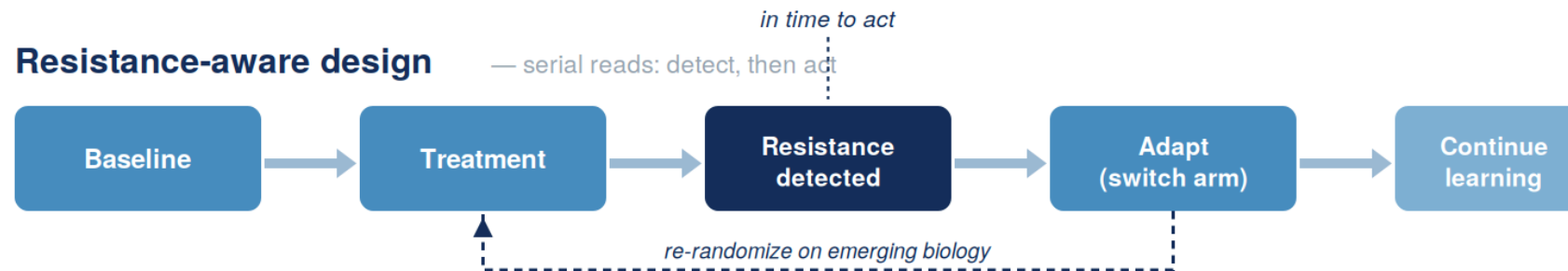
## Static design

— one biomarker read, before treatment



## Resistance-aware design

— serial reads: detect, then act



**A static read measures a moving target.**

Longitudinal detection + biomarker-guided arm-swaps keep the trial in step with the tumor.

# Detecting resistance takes two reads, not one

Different resistance mechanisms become visible through different assays.

| RESISTANCE TYPE                                                             | WHAT YOU NEED TO MEASURE IT                                  |
|-----------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Genomic</b><br>KRAS amplification, NRAS / BRAF / MAP2K1 alterations      | <b>ctDNA</b> <i>serial liquid biopsy, minimally invasive</i> |
| <b>Cell-state / plasticity</b><br>basal-to-classical switching, partial EMT | <b>biopsy + transcriptomics + imaging</b>                    |

**A resistance-aware trial has to run both, during enrollment, not at progression.**

*Genomic and cell-state resistance are invisible to each other's assay.*

# ADAPT / EVOLVE: what a resistance-aware trial looks like

## ARPA-H's resistance-tracking template.

*ARPA-H's flagship adaptive resistance-monitoring platform.*

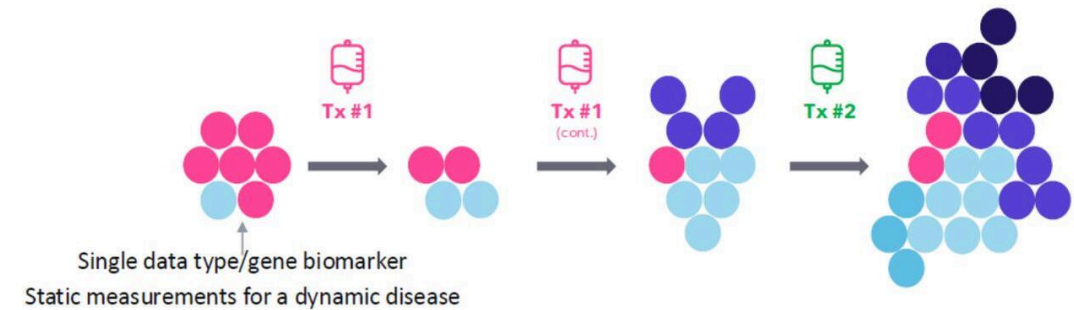
- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

*ARPA-H ADAPT → EVOLVE (UNC-led; Carey MPI, Rashid MPI). Enrolling.*

## ADAPT Goal: Identify and target changes in metastatic cancers

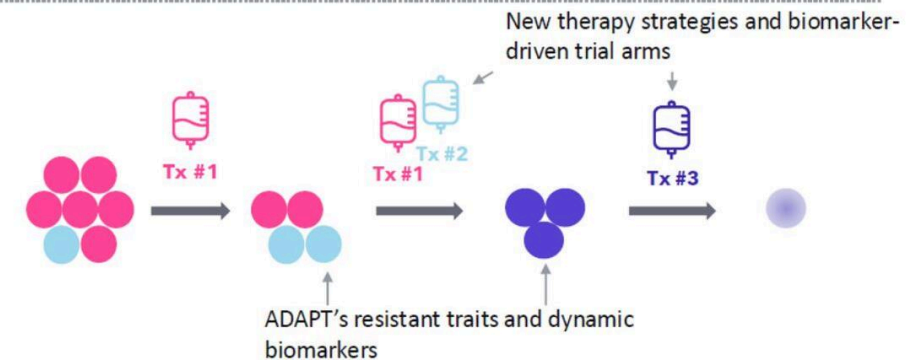
### Current State

- Cancer changes during treatment. Therapies **not** often informed by tumor change to cancer resistance traits.
- There are only **limited** tumor measurement biomarker analyses, which are most often performed **prior to** treatment.



### ADAPT's upcoming "future state"

- Therapies **will be** informed by and tailored to observed cancer resistance traits using biomarkers.
- **In-depth** tumor measurements will be taken **before, during and after** treatment.



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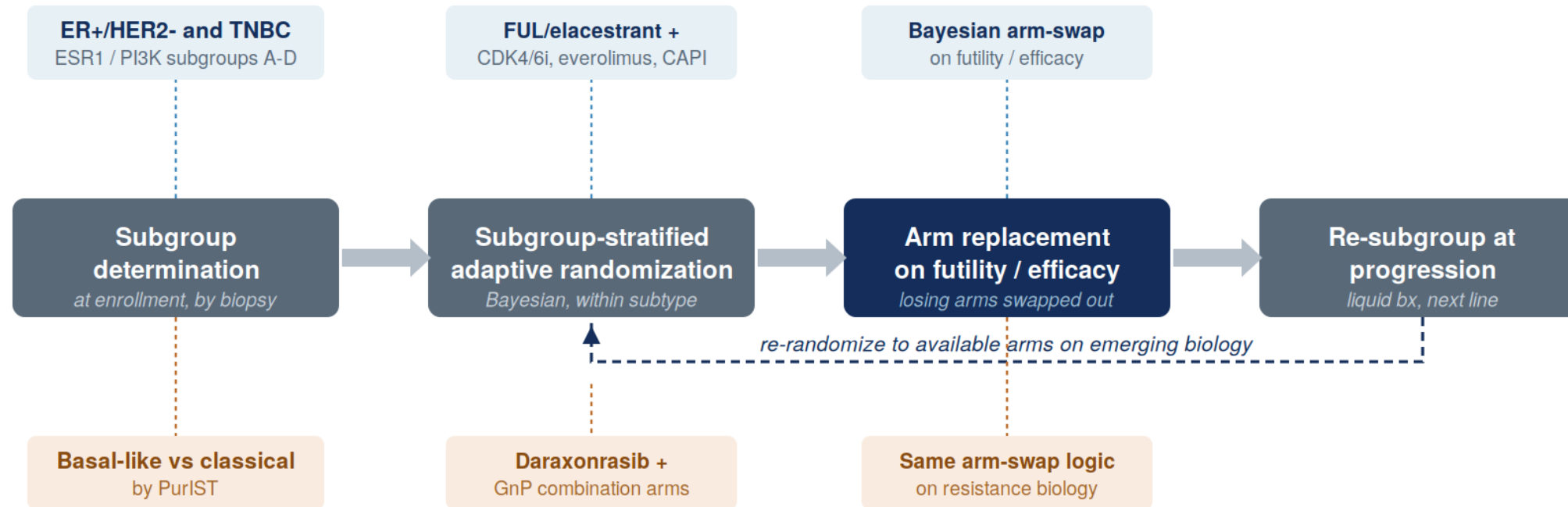
*ARPA-H ADAPT, current state (top) vs dynamic-biomarker state (bottom).*

# Same architecture, different disease

The EVOLVE-BDT flow is reusable: only the biomarker and the drugs change.

■ **EVOLVE-BDT — breast (real, enrolling)**

■ **PDAC analog (PurIST-defined, hypothetical)**



**The same adaptive framework applies to PurIST-defined PDAC subtypes and evolving resistance biology.**

Swap the biomarker and the drugs; the calibration machinery and decision rules carry over.

# What it would take to actually run these trials

Four things any such trial has to do at once.

**Stop ineffective  
therapies  
earlier**

Bayesian futility /  
efficacy boundaries  
at each interim

**Learn who benefits  
while the trial  
is running**

Pre-specified signatures  
+ agnostic  
discovery window

**Track resistance  
as it  
emerges**

Serial ctDNA,  
imaging, biopsies  
on every patient

**Test several  
biomarker groups  
under one protocol**

Biomarker-guided  
cohorts within the  
same protocol

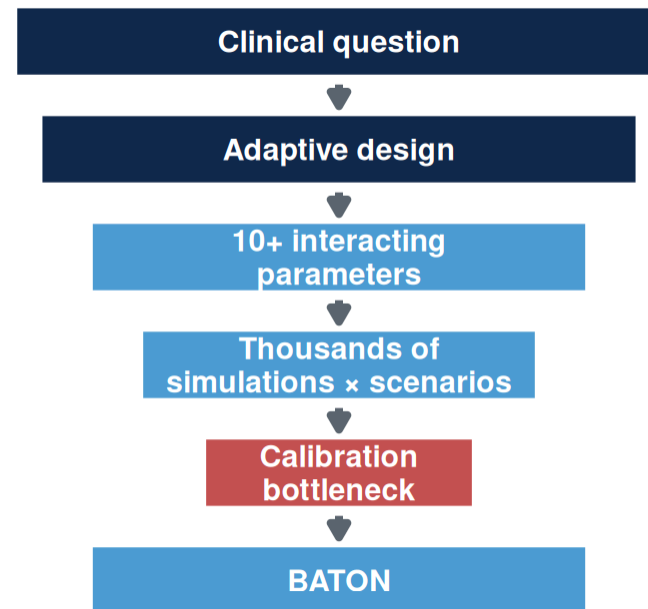
**Everything I've just described already exists statistically. So why don't we see these trials everywhere?**

# Putting these into practice

# Why don't we see more of these trials?

*Funding, accrual, and biopsy logistics are real. **But the barrier we can actually remove is calibration.***

Before an adaptive trial can open, **thousands of simulations** are needed to prove it is safe, efficient, and scientifically reliable. *That process, not the design itself, is often the bottleneck.*



Bayesian designs adopted 75%

Dose-optimization plans 30%

Our group: several months per design, hand-tuning quickly becomes infeasible.



# Calibration has to hit four goals at once

Designing an adaptive trial requires achieving all four simultaneously.

## 1. FDA compliance

Power  $\geq 0.80$   
Type I  $\leq 0.10$

## 2. Minimize patients

Smaller  $E[N]$   
Efficient use of  
scarce patients

## 3. Stop bad early

Futility stopping  
Protect patients  
from inactive arms

## 4. Accelerate good

Efficacy stopping  
Faster access  
when drug works

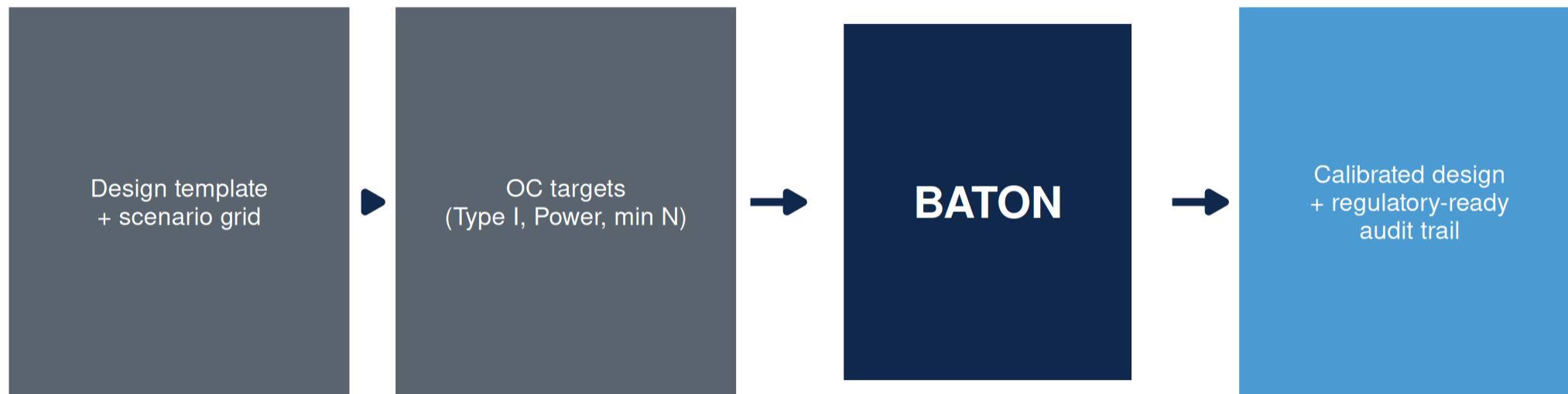
*Of thousands of plausible designs, only a small fraction satisfies all four, and you cannot find it by hand.*

***“We knew what trial to run.  
We needed a way to make it  
launchable.”***

# BATON: a way to make adaptive trials launchable

BATON automatically searches thousands of candidate designs and identifies those that meet FDA, efficiency, and scientific goals.

BATON = Bayesian Adaptive Trial Optimization<sup>23</sup>



# A BATON-calibrated design hits all four goals, for PANGEA Phase II

## 1. FDA compliance ✓

Type I 7.0% (target  $\leq 10\%$ )  
Power 82.2% (target  $\geq 80\%$ )

## 2. Minimize patients ✓

~23 patients spared  
(avg N 53 vs Max N 76)

## 3. Stop bad early ✓

78% early futility stop  
when drug is inactive

## 4. Accelerate good ✓

81% early efficacy stop  
when drug works

Far fewer evaluations than grid search, finds feasible designs grid search often misses entirely.

For this 4-parameter design: manual calibration takes days · BATON takes minutes - with a regulatory-ready audit trail.

# What this means for a PDAC patient

*Mr. Hernandez, 62, mPDAC, basal-like by PurlST, progressed on 1L GnP.*

## Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- **~36+ months** to read out

## PANGEA Phase II (BATON-calibrated)

- **78% chance** of early futility stop **when the drug is inactive**
- Average patients enrolled: **~53** if inactive / **~58** if active
- **27.3 months** expected duration

**Sooner answers. Fewer patients exposed to ineffective therapy. No sacrifice in statistical rigor.**

# PANGEA Phase II: calibrated by BATON

BATON identified a design that preserves power, controls false positives, and allows early stopping when erlotinib isn't helping.

- After Phase I OBD → **Phase II Bayesian adaptive RCT**: GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)<sup>24</sup>
- **BATON-calibrated** four parameters under Type I  $\leq 0.10$ , Power  $\geq 0.80$ <sup>23</sup>

Achieved operating characteristics:

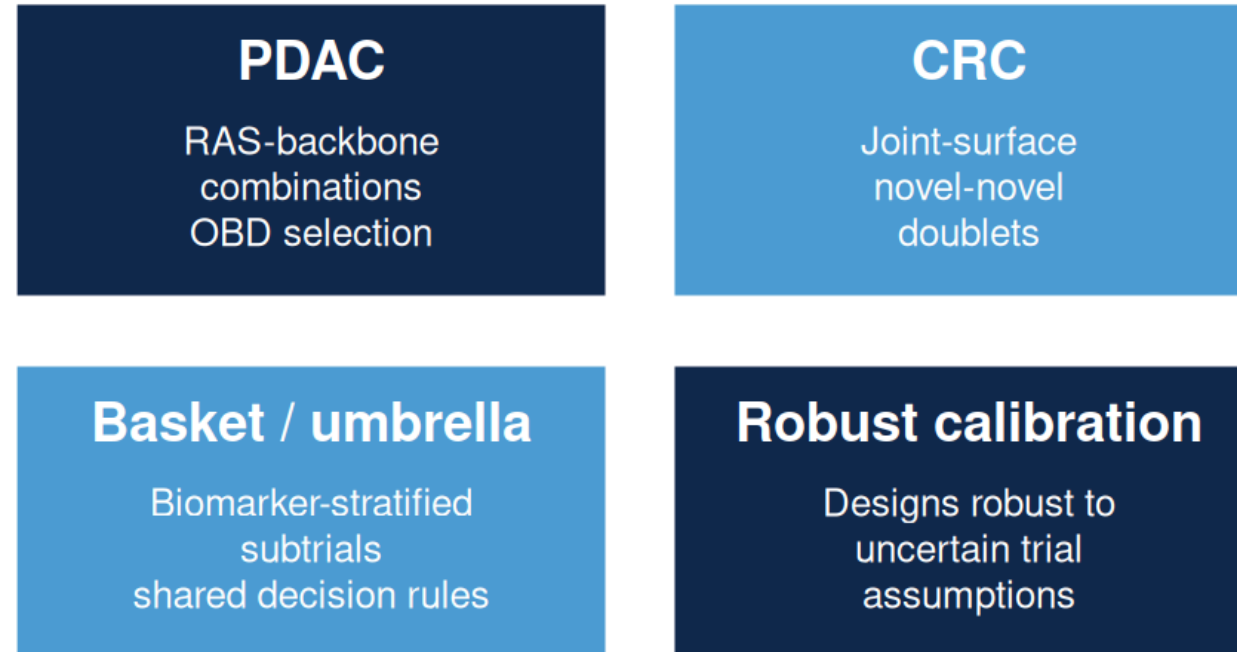
| METRIC                           | VALUE                     |
|----------------------------------|---------------------------|
| Power                            | 82.2%                     |
| Type I error                     | 7.0%                      |
| Max N                            | 76 (82 accrual w/ buffer) |
| Expected N under $H_0$           | 53.0                      |
| Expected N under $H_1$           | 58.1                      |
| Pr(early futility stop   $H_0$ ) | 77.9%                     |
| Pr(early efficacy stop   $H_1$ ) | 81.4%                     |
| Expected duration                | 27.3 mo                   |

$H_0$  = drug inactive ·  $H_1$  = drug active (HR 0.58). Read as: 78% early-stop when inactive; 81% early-stop when it works.

*Why max N = 76, not lower: caps set too low silently suppress early-efficacy stopping. BATON surfaced the trade-off curve; we chose 76 to preserve it.*

# What becomes possible

BATON calibrates **aggressive**, **conservative**, or **balanced** designs, and ports across GI applications:



**BATON is the infrastructure that makes adaptive trial design routine across the GI portfolio.**

*Adaptive, resistance-aware trials as the routine standard across GI, not the exception.*

# Take-home

1. **The RAS era needs new kinds of trials**, biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs can handle dose optimization, biomarker guidance, and resistance-aware adaptation.
3. **BATON** removes one of the major barriers preventing these designs from reaching patients.



# Acknowledgments

## Clinical collaborators

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- Ashwin Somasundaram (PANGAEA PI)

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- Charles Perou (UNC) · Ian Krop, Eric Winer (Yale) · Antonio Wolff (Hopkins)

## Methods collaborators

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# Questions